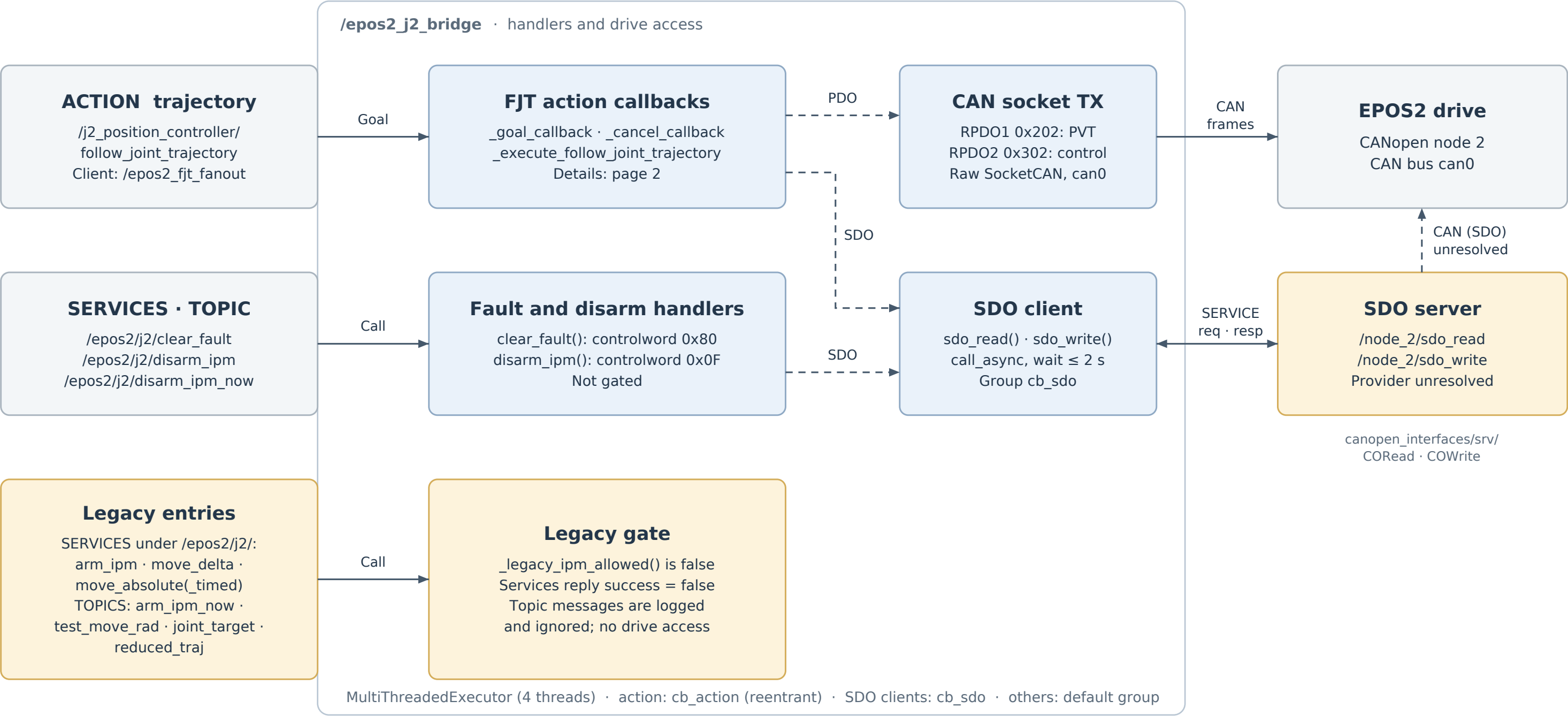


Level 2 — /epos2_j2_bridge entry points

epos2_bridge / epos2_joint_bridge · ROS node /epos2_j2_bridge · config/joints/j2.yaml · prefix /epos2/j2 · drive node 2 on can0
Source-derived · launched by epos2_j2_generic.launch.py · allow_legacy_ipm_services: false

Blue: node internals · Gray: ROS interfaces, peers, hardware · Amber: disabled or unresolved
Solid: ROS communication or CAN · Dashed: local call or shared state



Types: FollowJointTrajectory (control_msgs/action) · Trigger (std_srvs/srv) · MoveDelta, MoveAbsolute, MoveAbsoluteTimed (epos2_bridge_interfaces/srv)
disarm_ipm_now: std_msgs/msg/Bool · state, timers and telemetry: page 3

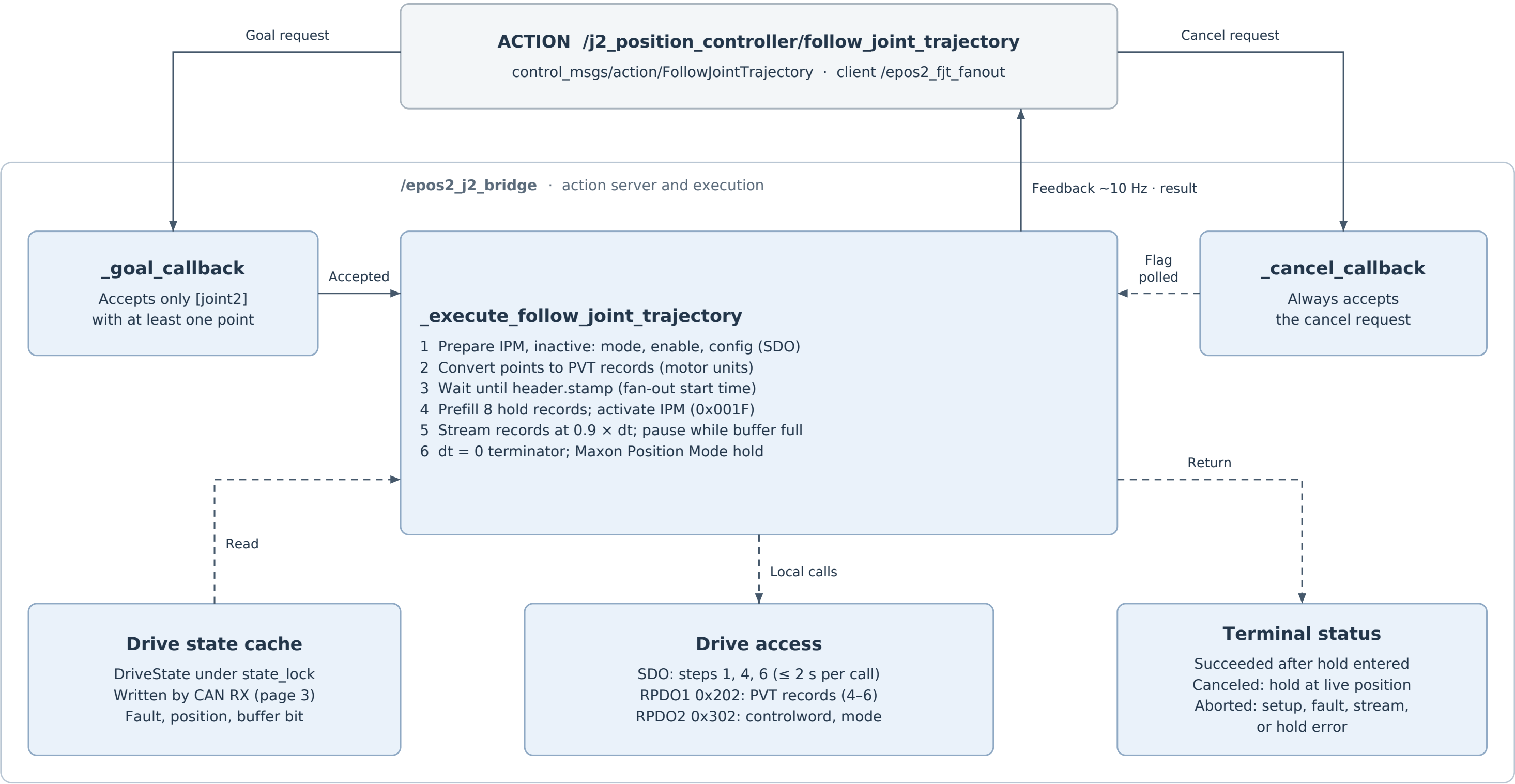
Source evidence

Agrobot/src/epos2_bridge/launch/epos2_j2_generic.launch.py:8-17 — epos2_joint_bridge named epos2_j2_bridge; config/joints/j2.yaml:3-21 — prefix, action, node 2, can0, legacy flag false.
Agrobot/src/epos2_bridge/epos2_bridge/epos2_joint_bridge.py:366-376 — SDO clients (cb_sdo) on /node_2/sdo_read|write; 378-387 — raw SocketCAN; 404-486 — subscriptions, action server (cb_action), services.
Same file 506-533, 1454-1480, 2423-2427 — clear_fault and disarm handlers (ungated); 870-890 — legacy gate; 517-527, 702-745, 2396-2421, 1987-1990, 1482-1484 — gated entries.
Same file 747-789 — SDO call with 2 s wait; 2156-2193 — RPDO2 and RPDO1 writers; 2745-2771 — MultiThreadedExecutor(num_threads=4).

Level 2 — /epos2_j2_bridge trajectory execution

Blue: node internals · Gray: ROS interfaces, peers, hardware · Amber: disabled or unresolved
Solid: ROS communication or CAN · Dashed: local call or shared state

epos2_bridge / epos2_joint_bridge · ROS node /epos2_j2_bridge · config/joints/j2.yaml · prefix /epos2/j2 · drive node 2 on can0
One-joint FollowJointTrajectory goal from /epos2_fjt_fanout · source-derived



Cancel is checked during the start-time wait and before each streamed record. A canceled goal returns error_code SUCCESSFUL.
cb_action is reentrant; blocking sleeps and SDO waits inside the execute callback occupy one of the 4 executor threads for the whole goal.

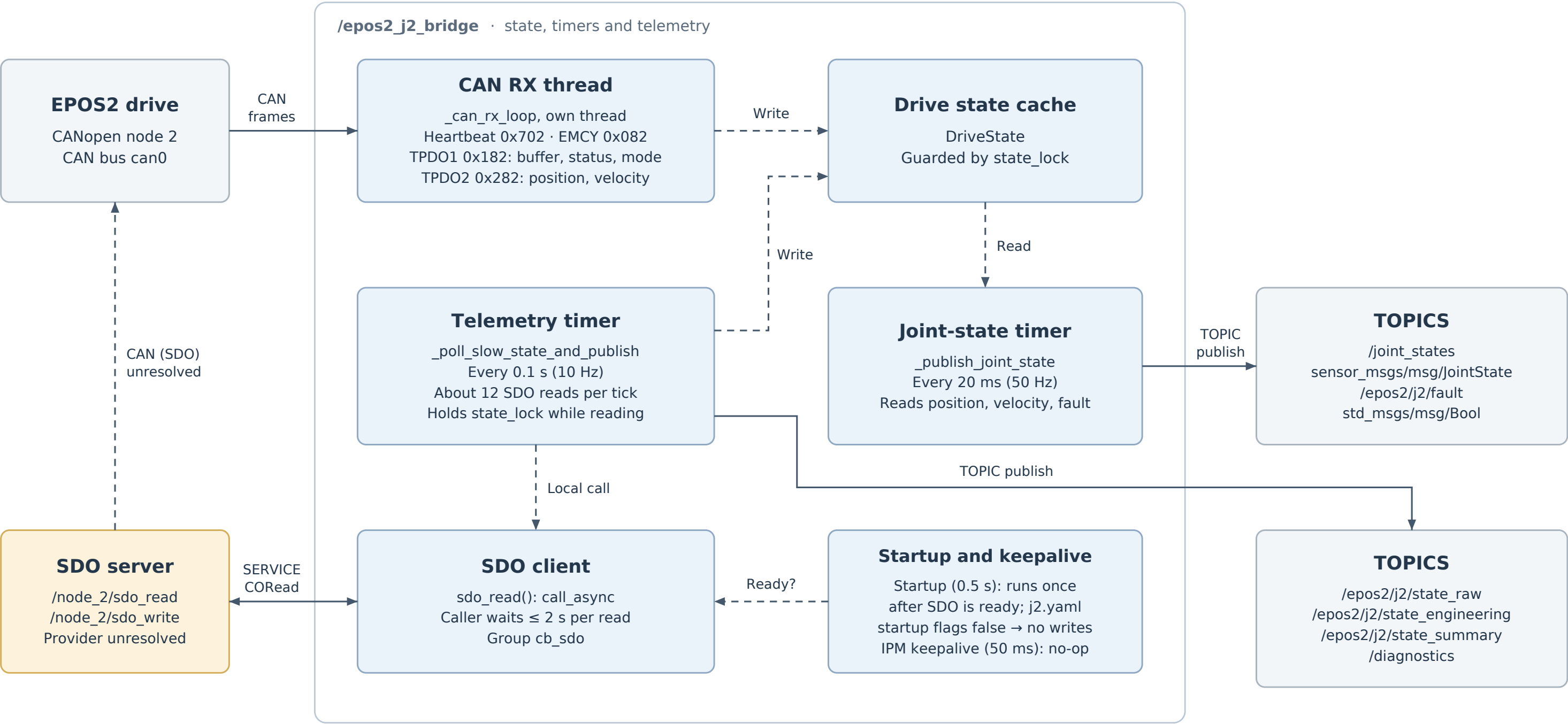
Source evidence

Agrobot/src/epos2_bridge/epos2_bridge/epos2_joint_bridge.py:438-447 — action server (cb_action); 2430-2443 — goal and cancel callbacks; 2445-2663 — execute callback.
Same file 2460 — prepare; 2477 — PVT conversion; 2498-2521 — start-time wait with cancel and fault checks; 2529-2538 — prefill and IPM enable.
Same file 2548-2628 — streaming, buffer-full pause, cancel to live-position hold, feedback; 2641-2662 — terminate, hold, succeed.
Same file 925-1004 — prepare_ipm_inactive; 1230-1324 — terminate_ipm_and_enter_position_hold; 2665-2682 — PVT conversion.

Level 2 — /epos2_j2_bridge state and telemetry

epos2_bridge / epos2_joint_bridge · ROS node /epos2_j2_bridge · config/joints/j2.yaml · prefix /epos2/j2 · drive node 2 on can0
Drive state inputs and periodic outputs · default rates (not overridden in j2.yaml) · source-derived

Blue: node internals · Gray: ROS interfaces, peers, hardware · Amber: disabled or unresolved
Solid: ROS communication or CAN · Dashed: local call or shared state



CAN RX runs in its own thread, outside the executor. Timers share the node's default mutually exclusive group, so a slow 10 Hz poll delays the 50 Hz publisher. While the poll holds state_lock, CAN RX updates, joint-state reads and trajectory-streaming state checks wait for it. Telemetry types: std_msgs/msg/Int64MultiArray, Float64MultiArray, String · diagnostic_msgs/msg/DiagnosticArray.

Source evidence

Agrobot/src/epos2_bridge/epos2_bridge/epos2_joint_bridge.py:378-387, 2201-2236 — raw SocketCAN RX thread; heartbeat, TPDO1, TPDO2 and EMCY decoded into DriveState. Same file 301-302, 487-503 — timers (50 Hz and 10 Hz defaults, 50 ms keepalive, 0.5 s startup); 2195-2199 — keepalive returns; 791-805 — startup. Same file 2239-2255 — /joint_states and fault; 2257-2346 — SDO polling inside state_lock, state_raw/engineering/summary; 2348-2394 — /diagnostics. Same file 366-376, 747-770 — SDO client in cb_sdo, 2 s wait; Agrobot/src/epos2_bridge/config/joints/j2.yaml:17-21 — startup flags false.